

**EXCEL ASSIGNMENT 2**  
**DATA CLEANING AND TRANSFORMATION**

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**Subject:** Data Analytics  
**Module:** Excel

## 1. MISSING VALUE HANDLING

**Explanation:** Missing values were identified in the Price and Category columns. The missing entries were handled to make the dataset complete and suitable for further analysis.

| Price (\$)  | Quantity | Category    |
|-------------|----------|-------------|
| \$ 1,000.00 | 30       | Electronics |
| \$ 80.00    | 15       | Fashion     |
| \$ 130.00   | 40       | Kitchen     |
| \$ 900.00   | 25       | Electronics |
| \$ 70.00    | 20       | Accessories |
|             | 45       | Electronics |
| \$ 30.00    | 5        | Fashion     |
| \$ 90.00    | 35       | Kitchen     |
| \$ 500.00   | 50       | Electronics |
| \$ 130.00   | 10       | Outdoor     |
| \$ 950.00   | 25       | Electronics |
| \$ 90.00    | 40       | Fashion     |
| \$ 120.00   | 35       | Kitchen     |
| \$ 150.00   | 15       | Electronics |
| \$ 250.00   | 20       | Electronics |
| \$ 50.00    | 35       | Accessories |
| \$ 180.00   | 15       | Electronics |
| \$ 980.00   | 10       | Electronics |
| \$ 150.00   | 15       | Fashion     |
|             | 10       | Outdoor     |
| \$ 700.00   | 50       | Electronics |
| \$ 80.00    | 20       | Kitchen     |
| \$ 150.00   | 30       | Accessories |
| \$ 800.00   | 45       | Electronics |
|             | 25       | Fashion     |
| \$ 400.00   | 40       | Kitchen     |
| \$ 60.00    | 30       | Fashion     |
| \$ 40.00    | 10       | Kitchen     |
| \$ 130.00   | 5        | Electronics |
| \$ 50.00    | 50       | Fashion     |
| \$ 100.00   | 20       | Accessories |

## 2.Data Standardization

| Product Name    | Brand Name | Product Brand           | Price (\$)  | Quantity | Category    |
|-----------------|------------|-------------------------|-------------|----------|-------------|
| Laptop          | Dell       | Dell Laptop             | \$ 1,000.00 | 30       | Electronics |
| Sneakers        | Nike       | Nike Sneakers           | \$ 80.00    | 15       | Fashion     |
| Coffee Maker    | Keurig     | Keurig Coffee Maker     | \$ 130.00   | 40       | Kitchen     |
| smartphone      | Samsung    | Samsung smartphone      | \$ 900.00   | 25       | Electronics |
| Backpack        | North Face | North Face Backpack     | \$ 70.00    | 20       | Accessories |
| Headphones      | Sony       | Sony Headphones         |             | 45       | Electronics |
| T-shirt         | Adidas     | Adidas T-shirt          | \$ 30.00    | 5        | Fashion     |
| Blender         | Ninja      | Ninja Blender           | \$ 90.00    | 35       | Kitchen     |
| Tablet          | Apple      | Apple Tablet            | \$ 500.00   | 50       | Electronics |
| Hiking Boots    | Timberland | Timberland Hiking Boots | \$ 130.00   | 10       | Outdoor     |
| Laptop          | HP         | HP Laptop               | \$ 950.00   | 25       | Electronics |
| Sneakers        | Adidas     | Adidas Sneakers         | \$ 90.00    | 40       | Fashion     |
| Coffee Maker    | Nespresso  | Nespresso Coffee Make   | \$ 120.00   | 35       | Kitchen     |
| Smartwatch      | Fitbit     | Fitbit Smartwatch       | \$ 150.00   | 15       | Electronics |
| Headphones      | Bose       | Bose Headphones         | \$ 250.00   | 20       | Electronics |
| Laptop Bag      | Samsonite  | Samsonite Laptop Bag    | \$ 50.00    | 35       | Accessories |
| Smartwatch      | Huawei     | Huawei Smartwatch       | \$ 180.00   | 15       | Electronics |
| Laptop          | Asus       | Asus Laptop             | \$ 980.00   | 10       | Electronics |
| Sunglasses      | Oakley     | Oakley Sunglasses       | \$ 150.00   | 15       | Fashion     |
| Camping Tent    | Coleman    | Coleman Camping Tent    |             | 10       | Outdoor     |
| Camera          | Nikon      | Nikon Camera            | \$ 700.00   | 50       | Electronics |
| Microwave       | Panasonic  | Panasonic Microwave     | \$ 80.00    | 20       | Kitchen     |
| Fitness Tracker | Xiaomi     | Xiaomi Fitness Tracker  | \$ 150.00   | 30       | Accessories |
| Smartphone      | Google     | Google Smartphone       | \$ 800.00   | 45       | Electronics |
| Sunglasses      | Ray-Ban    | Ray-Ban Sunglasses      |             | 25       | Fashion     |
| Blender         | Vitamix    | Vitamix Blender         | \$ 400.00   | 40       | Kitchen     |
| Dress           | Zara       | Zara Dress              | \$ 60.00    | 30       | Fashion     |
| Toaster         | Hamilton   | Hamilton Toaster        | \$ 40.00    | 10       | Kitchen     |
| Fitness Tracker | Garmin     | Garmin Fitness Tracker  | \$ 130.00   | 5        | Electronics |
| Jeans           | Levi's     | Levi's Jeans            | \$ 50.00    | 50       | Fashion     |
| Watch           | Casio      | Casio Watch             | \$ 100.00   | 20       | Accessories |

**Explanation:** Inconsistent text formats and category values were corrected using standardization and Find & Replace operations.

### 3.Duplicate Removal

| Product ID | Manufacturing Date | Country Code | Product Name | Brand Name | Product Brand           | Price (\$)  | Quantity | Category    |
|------------|--------------------|--------------|--------------|------------|-------------------------|-------------|----------|-------------|
| 28-JAN-US  | 28-01-2026         | US           | Laptop       | Dell       | Dell Laptop             | \$ 1,000.00 | 30       | Electronics |
| 15-FEB-US  | 15-02-2026         | US           | Sneakers     | Nike       | Nike Sneakers           | \$ 80.00    | 15       | Fashion     |
| 03-MAR-US  | 03-03-2026         | US           | Coffee Maker | Keurig     | Keurig Coffee Maker     | \$ 130.00   | 40       | Kitchen     |
| 11-APR-US  | 11-04-2026         | US           | smartphone   | Samsung    | Samsung smartphone      | \$ 900.00   | 25       | Electronics |
| 22-MAY-US  | 22-05-2026         | US           | Backpack     | North Face | North Face Backpack     | \$ 70.00    | 20       | Accessories |
| 07-JUN-UK  | 07-06-2026         | UK           | Headphones   | Sony       | Sony Headphones         |             | 45       | Electronics |
| 19-JUL-UK  | 19-07-2026         | UK           | T-shirt      | Adidas     | Adidas T-shirt          |             | 5        | Fashion     |
| 23-AUG-UK  | 23-08-2026         | UK           | Blender      | Ninja      | Ninja Blender           |             | 35       | Kitchen     |
| 05-SEP-UK  | 05-09-2026         | UK           | Tablet       | Apple      | Apple Tablet            |             | 50       | Electronics |
| 14-OCT-UK  | 14-10-2026         | UK           | Hiking Boots | Timberland | Timberland Hiking Boots |             | 10       | Outdoor     |
| 17-JUN-IN  | 17-06-2026         | IN           | Laptop       | HP         | HP Laptop               | \$ 950.00   | 25       | Electronics |
| 25-NOV-AU  | 25-11-2026         | AU           | Sneakers     | Adidas     | Adidas Sneakers         | \$ 90.00    | 40       | Fashion     |
| 08-DEC-DE  | 08-12-2026         | DE           | Coffee Maker | Nespresso  | Nespresso Coffee Maker  | \$ 120.00   | 35       | Kitchen     |
| 18-FEB-CA  | 18-02-2026         | CA           | Smartwatch   | Fitbit     | Fitbit Smartwatch       | \$ 150.00   | 15       | Electronics |
| 16-APR-ES  | 16-04-2026         | ES           | Headphones   | Bose       | Bose Headphones         | \$ 250.00   | 20       | Electronics |
| 21-AUG-CA  | 21-08-2026         | CA           | Laptop Bag   | Samsonite  | Samsonite Laptop Bag    | \$ 50.00    | 35       | Accessories |
| 20-AUG-CN  | 20-08-2026         | CN           | Smartwatch   | Huawei     | Huawei Smartwatch       | \$ 160.00   | 15       | Electronics |
| 27-JAN-IT  | 27-01-2026         | IT           | Laptop       | Asus       | Asus Laptop             | \$ 980.00   | 10       | Electronics |
| 01-MAR-UK  | 01-03-2026         | UK           | Sunglasses   | Oakley     | Oakley Sunglasses       | \$ 150.00   | 15       | Fashion     |
| 14-AUG-US  | 14-08-2026         | US           | Camping Tent | Coleman    | Coleman Camping Tent    |             | 10       | Outdoor     |
| 14-MAY-RU  | 14-05-2026         | RU           | Camera       | Nikon      | Nikon Camera            | \$ 700.00   | 50       | Electronics |

**Explanation:** Duplicate records were checked and removed from the dataset. The final dataset contains unique records.

## 4.Data Transformation

|    | A          | B                  | C            | D               | E          | F                      | G           | H        | I           |
|----|------------|--------------------|--------------|-----------------|------------|------------------------|-------------|----------|-------------|
| 1  | Product ID | Manufacturing Date | Country Code | Product Name    | Brand Name | Product Brand          | Price (\$)  | Quantity | Category    |
| 2  | 28-JAN-US  | 1/28/2026          | US           | Laptop          | Dell       | Dell Laptop            | \$ 1,000.00 | 30       | Electronics |
| 3  | 15-FEB-US  | 2/15/2026          | US           | Sneakers        | Nike       | Nike Sneakers          | \$ 80.00    | 15       | Fashion     |
| 4  | 03-MAR-US  | 3/3/2026           | US           | Coffee Maker    | Keurig     | Keurig Coffee Maker    | \$ 130.00   | 40       | Kitchen     |
| 5  | 11-APR-US  | 4/11/2026          | US           | smartphone      | Samsung    | Samsung smartphone     | \$ 900.00   | 25       | Electronics |
| 6  | 22-MAY-US  | 5/22/2026          | US           | Backpack        | North Face | North Face Backpack    | \$ 70.00    | 20       | Accessories |
| 7  | 07-JUN-UK  | 6/7/2026           | UK           | Headphones      | Sony       | Sony Headphones        |             | 45       | Electronics |
| 8  | 19-JUL-UK  | 7/19/2026          | UK           | T-shirt         | Adidas     | Adidas T-shirt         | \$ 30.00    | 5        | Fashion     |
| 9  | 23-AUG-UK  | 8/23/2026          | UK           | Blender         | Ninja      | Ninja Blender          | \$ 90.00    | 35       | Kitchen     |
| 10 | 05-SEP-UK  | 9/5/2026           | UK           | Tablet          | Apple      | Apple Tablet           | \$ 500.00   | 50       | Electronics |
| 11 | 14-OCT-UK  | 10/14/2026         | UK           | Hiking Boots    | Timberland | Timberland Hiking Boot | \$ 130.00   | 10       | Outdoor     |
| 12 | 17-JUN-IN  | 6/17/2026          | IN           | Laptop          | HP         | HP Laptop              | \$ 950.00   | 25       | Electronics |
| 13 | 25-NOV-AU  | 11/25/2026         | AU           | Sneakers        | Adidas     | Adidas Sneakers        | \$ 90.00    | 40       | Fashion     |
| 14 | 08-DEC-DE  | 12/8/2026          | DE           | Coffee Maker    | Nespresso  | Nespresso Coffee Mak   | \$ 120.00   | 35       | Kitchen     |
| 15 | 18-FEB-CA  | 2/18/2026          | CA           | Smartwatch      | Fitbit     | Fitbit Smartwatch      | \$ 150.00   | 15       | Electronics |
| 16 | 16-APR-ES  | 4/16/2026          | ES           | Headphones      | Bose       | Bose Headphones        | \$ 250.00   | 20       | Electronics |
| 17 | 21-AUG-CA  | 8/21/2026          | CA           | Laptop Bag      | Samsonite  | Samsonite Laptop Bag   | \$ 50.00    | 35       | Accessories |
| 18 | 20-AUG-CN  | 8/20/2026          | CN           | Smartwatch      | Huawei     | Huawei Smartwatch      | \$ 160.00   | 15       | Electronics |
| 19 | 27-JAN-IT  | 1/27/2026          | IT           | Laptop          | Asus       | Asus Laptop            | \$ 980.00   | 10       | Electronics |
| 20 | 01-MAR-UK  | 3/1/2026           | UK           | Sunglasses      | Oakley     | Oakley Sunglasses      | \$ 150.00   | 15       | Fashion     |
| 21 | 14-AUG-US  | 8/14/2026          | US           | Camping Tent    | Coleman    | Coleman Camping Tent   |             | 10       | Outdoor     |
| 22 | 14-MAY-RU  | 5/14/2026          | RU           | Camera          | Nikon      | Nikon Camera           | \$ 700.00   | 50       | Electronics |
| 23 | 09-JAN-CA  | 1/9/2026           | CA           | Microwave       | Panasonic  | Panasonic Microwave    | \$ 80.00    | 20       | Kitchen     |
| 24 | 19-JUL-BR  | 7/19/2026          | BR           | Fitness Tracker | Xiaomi     | Xiaomi Fitness Tracker | \$ 150.00   | 30       | Accessories |
| 25 | 29-SEP-CA  | 9/29/2026          | CA           | Smartphone      | Google     | Google Smartphone      | \$ 800.00   | 45       | Electronics |
| 26 | 03-JUN-CA  | 6/3/2026           | CA           | Sunglasses      | Ray-Ban    | Ray-Ban Sunglasses     |             | 25       | Fashion     |
| 27 | 11-JUL-CA  | 7/11/2026          | CA           | Blender         | Vitamix    | Vitamix Blender        | \$ 400.00   | 40       | Kitchen     |
| 28 | 07-MAR-CA  | 3/7/2026           | CA           | Dress           | Zara       | Zara Dress             | \$ 60.00    | 30       | Fashion     |
| 29 | 13-APR-CA  | 4/13/2026          | CA           | Toaster         | Hamilton   | Hamilton Toaster       | \$ 40.00    | 10       | Kitchen     |
| 30 | 24-MAY-CA  | 5/24/2026          | CA           | Fitness Tracker | Garmin     | Garmin Fitness Tracker | \$ 130.00   | 5        | Electronics |
| 31 | 02-DEC-CA  | 12/2/2026          | CA           | Jeans           | Levi's     | Levi's Jeans           | \$ 50.00    | 50       | Fashion     |
| 32 | 09-JUL-FR  | 7/9/2026           | FR           | Watch           | Casio      | Casio Watch            | \$ 100.00   | 20       | Accessories |
| 33 |            |                    |              |                 |            |                        |             |          |             |

### Explanation:

The Product ID column was split into Manufacturing Date and Country Code. The Brand Name and Product Name columns were merged into a new Product Brand column.

## 5.NUMBER FORMATTING AND CONDITIONAL FORMATTING

|      | Price (\$)  | Quantity | Category    |
|------|-------------|----------|-------------|
|      | \$ 1,000.00 | 30       | Electronics |
|      | \$ 80.00    | 15       | Fashion     |
| r    | \$ 130.00   | 40       | Kitchen     |
| ie   | \$ 900.00   | 25       | Electronics |
| ck   | \$ 70.00    | 20       | Accessories |
|      |             | 45       | Electronics |
|      | \$ 30.00    | 5        | Fashion     |
|      | \$ 90.00    | 35       | Kitchen     |
|      | \$ 500.00   | 50       | Electronics |
| ots  | \$ 130.00   | 10       | Outdoor     |
|      | \$ 950.00   | 25       | Electronics |
|      | \$ 90.00    | 40       | Fashion     |
| lake | \$ 120.00   | 35       | Kitchen     |
|      | \$ 150.00   | 15       | Electronics |
|      | \$ 250.00   | 20       | Electronics |
| ag   | \$ 50.00    | 35       | Accessories |
| i    | \$ 160.00   | 15       | Electronics |
|      | \$ 980.00   | 10       | Electronics |
|      | \$ 150.00   | 15       | Fashion     |
| ant  |             | 10       | Outdoor     |
|      | \$ 700.00   | 50       | Electronics |
| ve   | \$ 80.00    | 20       | Kitchen     |
| cer  | \$ 150.00   | 30       | Accessories |
|      | \$ 800.00   | 45       | Electronics |
| s    |             | 25       | Fashion     |
|      | \$ 400.00   | 40       | Kitchen     |

**Explanation:** The Price column was formatted as currency and data bars were applied to visually compare the prices. Cells containing the category “Electronics” were highlighted using conditional formatting.